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Date: _12/12/24_____ **Class:** 8

Element C

Design Requirement (HIGH Priority)

1. ADJUSTABILITY

There should be a function that allows our handle to adapt to people of most heights. The main problem with the cart in the first place was that someone of our stakeholder's height might have to lean over too much to push the cart and this led to back pains. Our product should make it easier for the stakeholder to just stand straight up and still be able to push the cart easily. Since this is the main goal of our product, this has a higher priority when compared with other requirements.

2. ERGONOMICALLY FIT

A cart handle should be ergonomic and feel comfortable while being used. Mr. Kiser, the stakeholder for this project, explained that he wanted the handle to feel somewhat comfortable and to have round grips with no sharp/straight edges. Even if our handle worked technically, if it was uncomfortable or if it hurt to hold, there would be no point in

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using the product. This may cause more pain, just to the hands instead of the back. We want our product to benefit our stakeholders as much as possible.

3. BUDGET

The product must be made using only thirty dollars or less. This is more of a restriction and not a requirement because our budget has been limited to thirty at a maximum so all of our supplies should be under that.

4. EASY REMOVABILITY

You should be able to put your handle on and take it off without tools or with minimal tools. During our interview, our stakeholder explained that if he could remove and insert the handle on the cart easily and without tools, this would make his experience with the product much better. If tools were needed, our client said that would be fine, but we still want to make sure that our product supports him while he is using the cart and even after.

5. LONG-LASTING

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Our cart handle must still be functional after at least five years of off and on usage. This design requirement came up because when we interviewed our stakeholder, Mr. Kiser, he explained that he wanted the handle to last until he left BHS, which would be in about five years.

Design Requirement (LOW Priority)

1. WIDE RANGE/UNIVERSAL

The handle should be able to attach to many types of carts so it is more universal. If we were to push this product out onto the market, our product should be compatible with all sorts of carts everywhere. One reason why this is a lower priority is because the design is supposed to be made for Mr. Kiser, our client, to push his already existing cart around the school. This cart might not have similar dimensions and characteristics as the cart we have been working with.

2. NO STORAGE INTRUSION

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Our product should try not to intrude on the storage space. Our stakeholders didn't want to give up too much area on the cart to have a better handle. This priority isn't as important as the rest because he said it is fine if some space is taken up, just not too much. For example, if our project takes away from a little bit of storage space on the bottom shelf, that would be fine. He just doesn't want to lose lots of space to store things.